CAPEX assessment

Supplementary material – **Combined Thermal Energy Storage and Dedicated Mechanical Subcooling for CO_2 refrigeration systems**

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1 Estimation of the CAPEX of a Transcritical CO_2 refrigeration cycle

The considered transcritical CO_2 refrigeration system comprises: a compressor, evaporators (closed refrigerated display cases) operating at a low temperature for useful cold production, a gas cooler ensuring heat rejection at ambient temperature, and an expansion valve. Since the economic contribution of the expansion valve is negligible compared to the total investment cost, only the first three components are considered in the CAPEX evaluation.

Cost of compressor

For the compressor, it is assumed that the correlations and assumptions reported in *Energy, exergy, and exergoeconomic analyses of an air source transcritical CO_2 heat pump for simultaneous domestic hot water and space heating* by A. Zendehboudi [1] are relevent, even though the CO_2 cycle is here used for a refrigeration system, rather than for a heat pump. It is assumed that the cost of the component varies little between the two.

According to this source, the investment cost of the compressor is given by:

$$Z_{\text{comp}} = 17547\,\dot{W}_{\text{comp}}^{0.4488} \quad [¥] \quad \Rightarrow \quad Z_{\text{comp}} = 2105\,\dot{W}_{\text{comp}}^{0.4488} \quad [¥]$$

where $\dot{W}_{\text{comp}}$ represents the compressor's absorbed electrical power in kW and with application of the current conversion rates (1¥ = 0.12€).

Cost of condenser/gas-cooler

One more time, a correlation developed by [1] is used. A corrective coefficient $k_{cor} = 1.5$ will then be applied to account for the fact that the intended use here is for refrigeration.

According to [1], the gas cooler is sized based on the number of plates N_{plate}, and its cost is given by:

$$Z_{\text{gc}} = 14.468\,N_{\text{plate}} + 411.12 \quad [€]$$

Heat exchange surface areas are evaluated by the relation:

$$A = \frac{Q}{U\,\Delta T_{\text{lm}}}$$

Considering the operating conditions of the fluid are adopted, relevent engineering values may be $U = 0.2$ kW m^{-2}K^{-1} and $\Delta T_{\text{lm}} = 10$ K for this gas-cooler.

Given the simulations that have already been generated, the representative value of $\text{COP}_{\text{CO}_2} = 3$ is considered to obtain:

$$\dot{Q}_{\text{gc}} = \dot{Q}_{\text{evap}} + \dot{W}_{\text{comp}}$$
$$\dot{Q}_{\text{gc}} = \dot{W}_{\text{comp}}(1 + \text{COP}_{\text{CO}_2})$$

Thus, assuming a unit area of $S = 0.25$ m^2 per plate, the number of plates is estimated as:

$$\begin{aligned} N_{\text{plate}} &= \frac{A_{\text{gc}}}{S} = \frac{\dot{Q}_{\text{gc}}}{S\,U\,\Delta T_{\text{lm}}} \\ &= \dot{W}_{\text{comp}} \frac{1 + \text{COP}_{\text{CO}_2}}{S\,U\,\Delta T_{\text{lm}}} \\ &= \dot{W}_{\text{comp}} \frac{1+3}{0.25 \cdot 10 \cdot 0.2} = 8 \cdot \dot{W}_{\text{comp}} \end{aligned}$$

Then the cost of the gas-cooler, prior to the correction k_{cor}, is given by:

$$Z_{\text{gc}} = 14.468 \cdot 8 \cdot \dot{W}_{\text{comp}} + 411.12$$
$$Z_{\text{gc}} = 115.744 \cdot \dot{W}_{\text{comp}} + 411.12 \quad [€]$$

Cost of evaporators

In the case of refrigeration for a supermarket, the evaporators are actually refrigerated display cases. The costs given by a supplier have been considered:

3636 €/kW for positive enclosed display cases

Where the power (kW) is the thermal cooling power $\dot{Q}_{\text{evap}}$. Thus it is possible to deduce the cost of this component:

$$\begin{aligned} \frac{Z_{\text{evap}}}{\dot{Q}_{\text{evap}}} &= 3636 \\ \Rightarrow Z_{\text{evap}} &= \text{COP}_{\text{CO}_2} \cdot \dot{W}_{\text{comp}} \cdot 3636 \\ \Rightarrow Z_{\text{evap}} &= 3 \cdot 3635 \cdot \dot{W}_{\text{comp}} \\ Z_{\text{evap}} &= 10908 \cdot \dot{W}_{\text{comp}} \ [€] \end{aligned}$$

Total CAPEX

The total investment cost is given by:

$$\begin{aligned} Z_{\text{tot}} &= Z_{\text{comp}} + Z_{\text{evap}} + k_{cor} \cdot Z_{\text{gc}} \\ &= (2105\,\dot{W}_{\text{comp}}^{0.4488}) + (10908 \cdot \dot{W}_{\text{comp}}) + 1.5 \cdot (115.744 \cdot \dot{W}_{\text{comp}} + 411.12) \end{aligned}$$

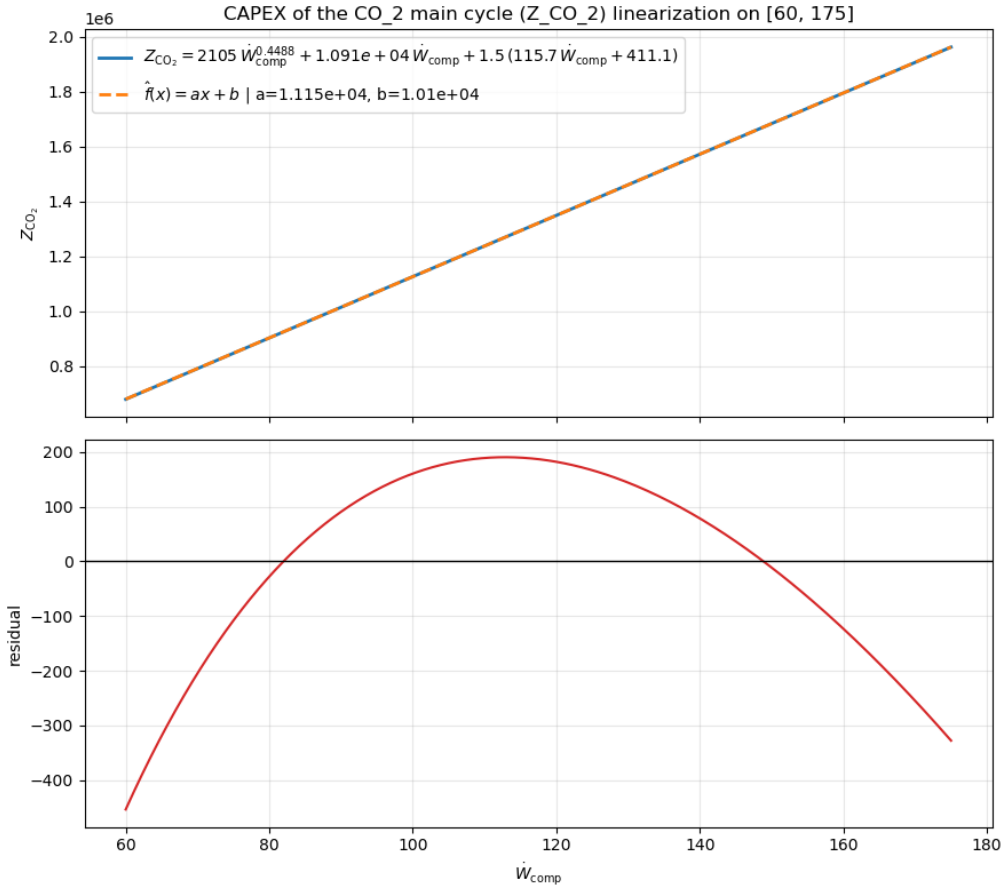

Figure 1: Linearization of $Z_{\text{tot CO}_2}$

It turns out that over the power range for which this correlation will be used (between 50 and 175 kW according to the simulations that have already been generated), the previous function can be approximated quite accurately by a linear function ($\mathbf{R^2} = 1$) as shown by the figure 1:

$$Z_{\text{tot CO}_2} \approx a\,\dot{W}_{\text{comp}} + b \quad \text{with} \quad a = 11151.4\ €/\text{kWe} \quad \text{and} \quad b = 10103\ €$$

Thus, the overall investment cost in € (where the basis is the compressor electrical power consumption) is:

$$\boxed{C_{\text{CO}_2} \approx 11151.4 \cdot \dot{W}_{\text{comp}} + 10103\ €}$$

2 Estimation of the CAPEX of a propane DMS System

This investment cost estimation methodology is based on correlations and assumptions reported in *Performance analysis and exergoeconomic evaluation of a TRC system enhanced by a dedicated*

mechanical subcooling by A. Z. Miran, A. Nemati and M. Yari (2019) [2].

The Dedicated Mechanical Subcooling (DMS) system investigated consists of the following components: a compressor, an evaporator providing thermal exchange with CO_2 (which is subcooled), an air-cooled condenser, and an expansion valve. Since the economic contribution of the expansion valve is negligible compared to the total investment cost, only the first three components are considered in the CAPEX evaluation.

Cost correlations from A. Z. Miran and al.

The research team uses different correlations that are here trusted. The current conversion rates (1\$ = 0.85€) are applied to convert \$ in €.

The purchase cost of heat exchangers:

$$Z_k = Z_{R,k} \left(\frac{A_k}{A_R} \right)^{0.6} \quad [\$]$$

where $Z_{R,k}$ is the reference cost, $A_R = 100\ \mathrm{m}^2$ is the reference heat transfer area, and A_k is the heat transfer area of the exchanger under consideration. Since the evaporator of the R290 is a subccoler for the R744 of the main cycle, the reference costs provided are:

$$Z_{R,\mathrm{evap}} = 1200\ \$ = 1020$$
$$Z_{R,\mathrm{cond}} = 8000\ \$ = 6800$$

The purchase cost of compressor:

$$Z_{comp} = 10167.5 \cdot \dot{W}_{comp}^{0.46}\ [\$]$$
$$Z_{comp} = 8642.4 \cdot \dot{W}_{comp}^{0.46}\ [€]$$

where $\dot{W}_C$ is the compressor electrical power consumption expressed in kW.

Simplified thermal sizing of heat exchangers

Given the simulations that have already been generated, the representative value of $\mathrm{COP}_{\mathrm{DMS}} = 5.5$ is considered to obtain:

$$\begin{cases} \dot{Q}_{\mathrm{evap}} & = \dot{W}_{\mathrm{comp}} \cdot \mathrm{COP}_{\mathrm{DMS}} \\ \dot{Q}_{\mathrm{cond}} & = \dot{Q}_{\mathrm{evap}} + \dot{W}_{\mathrm{comp}} = \dot{W}_{\mathrm{comp}}(1 + \mathrm{COP}_{\mathrm{DMS}}) \end{cases}$$

Then, the heat transfer surface area is evaluated as:

$$A = \frac{Q}{U\,\Delta T_{\mathrm{lm}}}$$

Typical engineering values are assumed for the overall heat transfer coefficient U and the logarithmic mean temperature difference ΔT_{lm}, considering the operating fluid conditions:

- R290/CO_2 evaporator: $U_{evap} = 0.8\ \mathrm{kW\ m^{-2}K^{-1}}$ and $\Delta T_{\mathrm{lm}\ evap} = 5\ \mathrm{K}$

- R290/air condenser: $U_{cond} = 0.05$ kW m^{-2}K^{-1} and $\Delta T_{\text{lm } cond} = 10$ K

Thus it is possible to express:

$$
\begin{aligned}
Z_k &= Z_{R,k} \left(\frac{Q}{A_R\, U\, \Delta T_{\text{lm}}} \right)^{0.6} \\
&\Rightarrow \begin{cases} Z_{evap} = Z_{R,evap} \left(\frac{\dot{W}_{\text{comp}} \cdot \text{COP}_{\text{DMS}}}{A_R\, U_{evap}\, \Delta T_{\text{lm } evap}} \right)^{0.6} \\ Z_{cond} = Z_{R,cond} \left(\frac{\dot{W}_{\text{comp}} (1+\text{COP}_{\text{DMS}})}{A_R\, U_{cond}\, \Delta T_{\text{lm } cond}} \right)^{0.6} \end{cases} \\
&\Rightarrow \begin{cases} Z_{evap} = \dot{W}_{\text{comp}}^{0.6} \cdot Z_{R,evap} \left(\frac{\text{COP}_{\text{DMS}}}{A_R\, U_{evap}\, \Delta T_{\text{lm } evap}} \right)^{0.6} \\ Z_{cond} = \dot{W}_{\text{comp}}^{0.6} \cdot Z_{R,cond} \left(\frac{1+\text{COP}_{\text{DMS}}}{A_R\, U_{cond}\, \Delta T_{\text{lm } cond}} \right)^{0.6} \end{cases} \\
&\Rightarrow \begin{cases} Z_{evap} = \dot{W}_{\text{comp}}^{0.6} \cdot 1020 \left(\frac{5.5}{100 \cdot 0.8 \cdot 5} \right)^{0.6} = 77.91 \cdot \dot{W}_{\text{comp}}^{0.6} \\ Z_{cond} = \dot{W}_{\text{comp}}^{0.6} \cdot 6800 \left(\frac{1+5.5}{100 \cdot 0.05 \cdot 10} \right)^{0.6} = 1999 \cdot \dot{W}_{\text{comp}}^{0.6} \end{cases}
\end{aligned}
$$

Total CAPEX

The total investment cost is given by:

$$
\begin{aligned}
Z_{\text{tot}} &= Z_{\text{comp}} + Z_{\text{evap}} + Z_{\text{cond}} \\
&= (8642.4 \cdot \dot{W}_{comp}^{0.46}) + (77.91 \cdot \dot{W}_{\text{comp}}^{0.6}) + (1999 \cdot \dot{W}_{\text{comp}}^{0.6})
\end{aligned}
$$

It turns out that over the power range for which this correlation will be used (between 4 and 12 kW according to the simulations that have already been generated), the previous function can be approximated quite accurately by a linear function ($\mathbf{R^2} = 0.995$) as shown by the figure 2:

$$
Z_{\text{tot DMS}} \approx a\, \dot{W}_{\text{comp}} + b \quad \text{with} \quad a = 1873.6 \text{ €/kWe} \quad \text{and} \quad b = 14413 \text{ €}
$$

Thus, the overall investment cost in € (where the basis is the compressor electrical power consumption) is:

$$
\boxed{C_{\text{DMS}} \approx 1873.6 \cdot \dot{W}_{\text{comp}} + 14413 \text{ €}}
$$

3 Estimation of the CAPEX of a TES System

Here, we use data collected by Celsiuscity-eu , a European project for disseminating and sharing experiences in the development of smart heating and cooling networks. In their 2015 report, Thermal Energy Storage they list recent thermal energy storage projects and their associated investment costs.

The graph on page 12 of the PDF shows the investment cost per m^3 water equivalent [€/m^3] as a function of the storage volume in water equivalent. Tanks for latent energy storage (like the project considered here) are represented by triangles. For our volume range (less than 100 m^3), the average price of 460€/m^3 is considered, where the volume is the volume of "water equivalent" that is v_{TES} (the volume of one of the 2 tanks of the TES system implemented in this study).

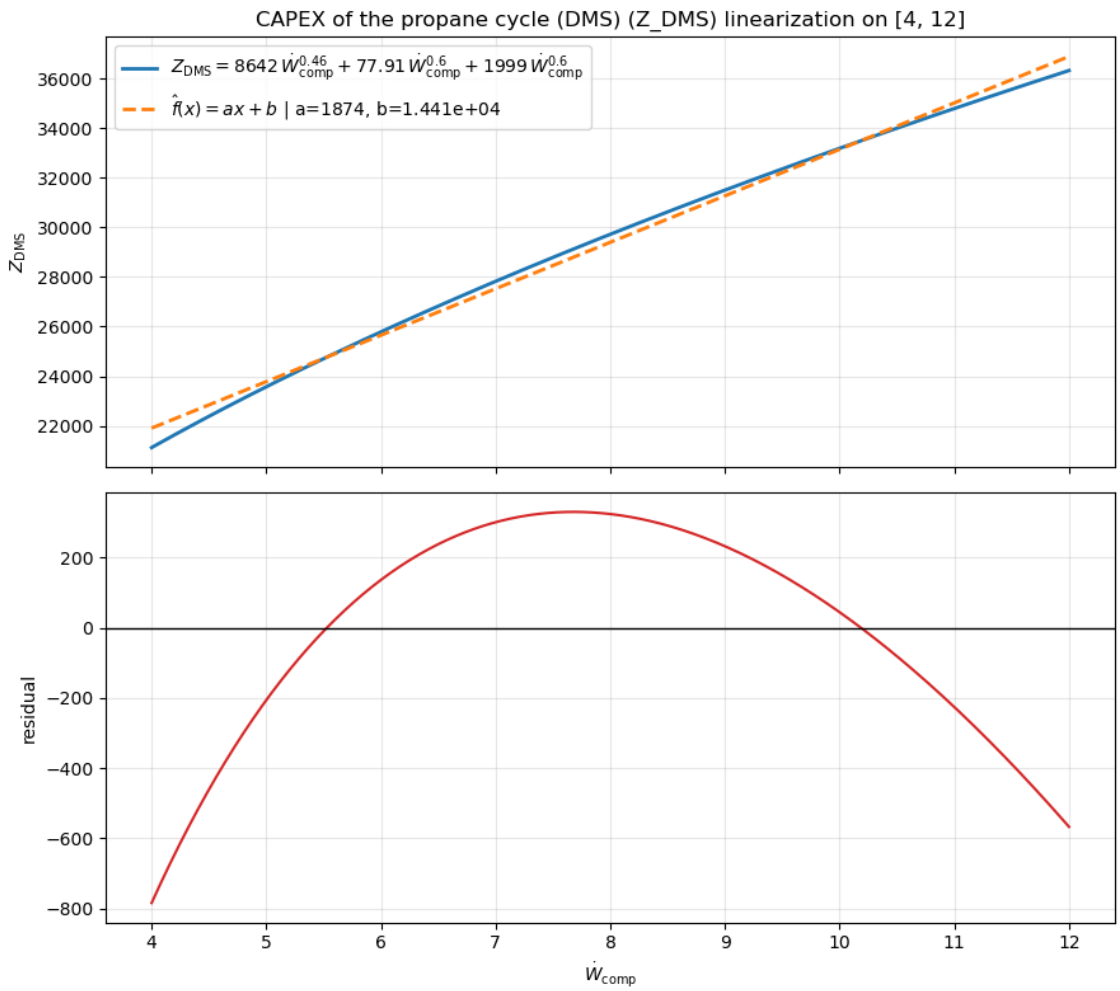

Figure 2: Linearization of $Z_{\mathrm{tot\ CO_2}}$

According to the document: *"The estimated cost of the thermal energy systems considers storage material, operation costs, and technical equipment for charging and discharging the storage device."* However, the cost of the additional heat exchanger required for propane recharging (not otherwise taken into account) must be added to this value. For this component, the price is close to that of the reference *Freon Heat Exchanger Nordic Tec Ba-68-20-F 1.36m²* (Refrigerant-to-Water Plate Heat Exchanger available on Nordictec-store.com, which is 335€ for approximately 10 to 14 kW capacity.

Ultimately, to account for the fact that the storage system in this study uses not one but two tanks, and also because the heat exchanger for discharge between the hydraulic system and the main cycle must be added, the value of the specific cost of the TES obtained using the report from *Celsiuscity-eu* is increased. Thus, the considered overall investment cost in €/m^3 (where the basis is v_{TES} the volume of each water tank, which are the same) is:

$$\boxed{c_{\text{TES}} \approx 800 \text{ €/m}^{\mathbf{3}}}$$

References

[1] A. Zendehboudi, *Energy, exergy, and exergoeconomic analyses of an air source transcritical CO_2 heat pump for simultaneous domestic hot water and space heating*, Energy, vol. 290, p. 130295, 1 Mar. 2024. DOI: 10.1016/j.energy.2024.130295.

[2] A. Z. Miran, A. Nemati, and M. Yari, *Performance analysis and exergoeconomic evaluation of a TRC system enhanced by a dedicated mechanical subcooling*, Energy Conversion and Management, vol. 199, p. 111890, 1 Nov. 2019. DOI: 10.1016/j.enconman.2019.111890.